

BullSO 2026 Invitational

Circuit Lab B Answer Key

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|---------|----------|----------|
| 1. B | 13. D | 25. C |
| 2. C | 14. C | 26. D |
| 3. C | 15. A | 27. D |
| 4. D | 16. B | 28. B |
| 5. B | 17. B, D | 29. B, C |
| 6. A | 18. B | 30. A |
| 7. A, D | 19. D | 31. B |
| 8. C | 20. D | 32. C |
| 9. D | 21. B | 33. C |
| 10. B | 22. A | 34. B |
| 11. A | 23. D | |
| 12. B | 24. C | |

PART 2: Short Answer

1.

- Resistor
- Capacitor
- Battery
- Diode
- Switch or open switch

2.

- 0.0018 Amps (A) or 1.8 milliamps (mA)
- 0.0162 Watts (W) or 16.2 milliwatts (mW)
- 3000 Ohms or 3k Ohms

3. 5.5 Ohms

4. 27 Ohms

5.

- Kirchoff's Current Law (KCL)
- Ohm's Law
- $I_1 = 2.625$ Amps (A), $I_2 = 0.625$ Amps (A)
 - 1 pt if units are missing/incorrect

6. $V_1 = 14.4$ Volts (V), $V_2 = 17.6$ Volts (V)

- +1 pt for correct KVL equation (may be written with Ohm's Law substituted in)
- +1 pt for correct current value
- +1 pt for **each** correct final voltage value
- Max final score of 3 pts if one/both final voltage values have missing/incorrect units
- Example Solution:

$$\text{KVL: } -20 + V_1 - 12 + V_2 = 0 \rightarrow V_1 + V_2 = 32$$

$$\text{Ohm's Law: } I(22) + I(18) = 32 \rightarrow 40I = 32 \rightarrow I = 0.8 \text{ Amps (A)}$$

$$V_1 = 0.8(18) = 14.4 \text{ Volts (V)}, V_2 = 0.8(22) = 17.6 \text{ Volts (V)}$$

7. $V_1 = 28 \text{ Volts (V)}$, $V_2 = 30 \text{ Volts (V)}$

- +1 pt for **each** correct KCL loop equation
- +1 pt for **each** correct final node voltage value
- Max final score of 3 pts if one/both final node voltage values have missing/incorrect units
- Example Solution:

$$\text{KCL Node 1: } \frac{V_1}{4} - \frac{V_2 - V_1}{2} = 6 \rightarrow 3V_1 - 2V_2 = 24$$

$$\text{KCL Node 2: } \frac{V_2}{6} + \frac{V_2 - V_1}{2} = 6 \rightarrow -3V_1 + 4V_2 = 36$$

Solve for V_1 , V_2

8. $I_1 = 5 \text{ Amps (A)}$, $I_2 = 2.5 \text{ Amps (A)}$

- +1 pt for **each** correct KVL loop equation
- +1 pt for **each** correct final current value
- Max final score of 3 pts if one/both final current values have missing/incorrect units
- Example Solution:

$$\text{KVL Loop 1: } -25 + 4I_1 + 2(I_1 - I_2) = 0 \rightarrow 6I_1 - 2I_2 = 25$$

$$\text{KVL Loop 2: } -15 + 8I_2 + 2(I_2 - I_1) = 0 \rightarrow -2I_1 + 10I_2 = 25$$

Solve for I_1 , I_2

9. Any 5 of the following are acceptable

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|--|--|
| • Infinite Open-Loop Gain (G) | • Infinite Power Supply Reject Ratio |
| • Infinite Input Impedance (R_{in}) | • Infinite Bandwidth |
| • Infinite Out Voltage Range ($v_{out \text{ max}}$) | • Zero Input Offset Voltage |
| • Infinite Common-Mode Rejection Ratio | • Zero Output Impedance (R_{out}) |
| | • Zero Noise |
| | • Zero Input Current (v_{in}/R_{in}) |

Lab - answers 24 pts - 5 extra points for B given for the total and explanation

A clearly labeled sketch of your circuit. - 8 pts

Make sure they label resistor values - 3pts. Use parallel - 1.5pts. Use series - 1.5 pts.

Proper diagram - resistor images/wire connections - 1 pts. And battery/complete circuit - 1pt

Step-by-step calculations showing intermediate equivalent resistances - 8 pts

2 pts per calculation for at least 3. 2pts for final calculation

A short explanation of why your configuration produces 133.333Ω . - 8 pts [13pts for B]

Make sure their total is 133.333 - 6 points [9 for B]. 2 points for explanation [4 for B].

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Example { series 100, 3 way parallel of 300, 50, and 150 }